

TD 1 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ pwd
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ ls
CONTRIBUTING.md  TEAM.md          config  evidence  readiness  src
README.md        TEAM_TEMPLATE.md data    labs      requirements.txt
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ ls -la
total 68
drwxrwxrwx+ 9 codespace root  4096 Oct  7 13:30 .
drwxr-xrwx+ 5 codespace root  4096 Oct  7 13:30 ..
drwxrwxrwx+ 8 codespace root  4096 Oct  7 13:31 .git
-rw-rw-rw-  1 codespace root   178 Oct  7 13:30 .gitignore
-rw-rw-rw-  1 codespace root  2943 Oct  7 13:30 CONTRIBUTING.md
-rw-rw-rw-  1 codespace root 10270 Oct  7 13:30 README.md
-rw-rw-rw-  1 codespace root  1429 Oct  7 13:30 TEAM.md
-rw-rw-rw-  1 codespace root  1446 Oct  7 13:30 TEAM_TEMPLATE.md
drwxrwxrwx+ 2 codespace root  4096 Oct  7 13:30 config
drwxrwxrwx+ 3 codespace root  4096 Oct  7 13:30 data
drwxrwxrwx+ 2 codespace root  4096 Oct  7 13:30 evidence
drwxrwxrwx+ 2 codespace root  4096 Oct  7 13:30 labs
drwxrwxrwx+ 2 codespace root  4096 Oct  7 13:30 readiness
-rw-rw-rw-  1 codespace root   133 Oct  7 13:30 requirements.txt
drwxrwxrwx+ 2 codespace root  4096 Oct  7 13:30 src
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ cd data
bash: cd: data: No such file or directory
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data/sample
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ ls
bloomberg_reference_expected.json  bloomberg_reference_sample.json  instruments.json  prices.csv
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ cd sample
bash: cd: sample: No such file or directory
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data/sample
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ ls
bloomberg_reference_expected.json  bloomberg_reference_sample.json  instruments.json  prices.csv
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07/data/sample (ayseguneysu) $ cd ../../
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ pwd
/workspaces/esilv-marketpulse-g01-t07
```

Part 6 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

The instrument is **Apple Inc. (AAPL)**, traded on the NASDAQ.

The benchmark is the **S&P 500 (SP500)**.

Their names and tickers are stored in `data/sample/instruments.json`, under the instrument and benchmark sections.

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

Part 7

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python --version
Python 3.14.2
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ git --version
git version 2.55.0
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```

Optional 1 :

Having two aligned series is useful because both the stock and the benchmark have data for the same dates. This makes it easier to compare their performance over time.

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ grep -c AAPL data/sample/prices.csv
21
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ grep -c SP500 data/sample/prices.csv
21
```

Optional 2 :

`ls -lh` shows the files and folders in the current directory. `head -n 5` shows the first 5 lines of the CSV, while `tail -n 5` shows the last 5 lines.

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ ls -lh
total 52K
-rw-rw-rw- 1 codespace root 2.9K Oct  7 13:30 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 11K Oct  7 13:30 README.md
-rw-rw-rw- 1 codespace root 1.4K Oct  7 13:30 TEAM.md
-rw-rw-rw- 1 codespace root 1.5K Oct  7 13:30 TEAM_TEMPLATE.md
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 config
drwxrwxrwx+ 3 codespace root 4.0K Oct  7 13:30 data
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 evidence
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 labs
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 readiness
-rw-rw-rw- 1 codespace root 133 Oct  7 13:30 requirements.txt
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:30 src
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ head -n 5 data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ tail -n 5 data/sample/prices.csv
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
```

Td 2 :

Part 1 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ code src/main.py
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```

```

from pathlib import Path
import csv
import json

DATA_DIR = Path("data/sample")

# These starter values mirror config/settings.yml.
# settings.yml is a human-readable configuration contract in the CORE.
# Parsing YAML is optional and is not required by the 18-hour lab sequence.
LOOKBACK_LABEL = "1 month"
INTERVAL_LABEL = "Daily"

def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)

def load_prices():
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]

if __name__ == "__main__":
    main()

```

- `load_instruments()` reads the JSON file.
- `load_prices()` reads the CSV file.
- `filter_prices()` selects the data for one ticker.
- `main()` runs and coordinates everything.
- The input files are in the `data/sample/` folder.

Part 2 :

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
```

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    print(type(instruments))
    print(instruments.keys())

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    instrument_latest = instrument_prices[-1]
    benchmark_latest = benchmark_prices[-1]

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    print(instrument["ticker"])
    print(instrument["name"])
    print(instrument["currency"])

    print(benchmark["ticker"])
    print(benchmark["name"])
```

Part 3 :

```
def main():
    instruments = load_instruments()
    prices = load_prices()

    print(type(prices))
    print(type(prices[0]))
    print(prices[0])
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneysu) $ python src/main.py
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
```

```
IndentationError: unindent does not match any outer indentation level
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ python src/main.py  
250.00  
<class 'str'>
```

```
def main():  
    instruments = load_instruments()  
    prices = load_prices()  
  
    print(prices[0]["close"])  
    print(type(prices[0]["close"]))
```

Part 4 :

```
def main():  
    instruments = load_instruments()  
    prices = load_prices()  
  
    instrument = instruments["instrument"]  
    benchmark = instruments["benchmark"]  
  
    instrument_prices = filter_prices(  
        prices,  
        instrument["ticker"],  
    )  
  
    benchmark_prices = filter_prices(  
        prices,  
        benchmark["ticker"],  
    )  
  
    print(len(instrument_prices))  
    print(len(benchmark_prices))
```

```
@ayse3993 →/workspaces/esilv-marketpulse-g01-t07 (ayseguneyasu) $ python src/main.py  
21  
21
```

Part 5 :